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Overview of alternative fuels with emphasis on the potential of liquefied natural gas as future marine fuel

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Abstract

Economic and population growths are the most important drivers of growing global energy demand. They led to a rapid development of international seaborne trade and an increase in the number of global vessels. Air pollution from these ships is of great concerns and regulations are currently enforced since May 2005 by the International Maritime Organization to limit such pollution. In this study, we will first review the current global energy demand and its driving forces over next decades, second evaluate the existing alternative fuels that can be used as a bunker fuel to reach sustainability with relatively small changes in the existing marine propulsion options and finally focus on near-term solution, which has the potential for large-scale use. The different alternative fuels were compared in terms of several parameters such as availability, renewability, safety, cost, performance, economy and compliance with emission regulations. This comparison revealed that liquefied natural gas could be considered as the future replacement to the current marine bunker fuel. This conclusion has been further verified by comparing diesel engine with different powers when using both heavy fuel oil and liquefied natural gas. The engines were compared against fuel consumption, cost saving as well as emissions. Liquefied natural gas has proved to be better than heavy fuel oil due to fuel cost reduction by about 31% per year and decrease in emissions of SO_x, NO_x, CO_2 and particulate matter by about 98%, 86%, 11% and 96%, respectively. The resulted emissions from using liquefied natural gas were found to comply with the current International Maritime Organization regulations. Moreover, this article highlights the latest rules and regulations that govern the use of liquefied natural gas as marine fuel onboard ships.

Keywords

Alternative fuels, natural gas, eco-friendly fuel, gas emission, marine engineering

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Introduction

Efforts to reduce costs and to achieve maximal profits in sea transport business are carried out to counter the steady increase in prices of oil and petrochemical products. Most of these efforts are aimed on research and development (R&D) to introduce new technologies in shipbuilding and other related maritime industries. In general, development is directed on high efficiency propulsion and maneuvering systems, advanced hull forms and alternative fuel and propulsion.[1] The conventional marine fuels have worked efficiently through the last decades, especially with regard to adaptability, performance and safety. But during the last few years, and due to the high quantity of emissions emitted from ships, strict emission regulations have been introduced by International Maritime Organization (IMO). Several inventory studies suggested that in 2000, ocean-going ships have emitted around 600–900 Tg CO_2, 15% of all global NO_x emissions and 4%–9% of global SO_2 emissions.[2] While in 2007, the quantity of gases emitted from ships was estimated to be 25 and 15 million tons of NO_x and SO_x, respectively, and around 2.7% of all global CO_2 are attributable to ships.[3] Other studies revealed that shipping-related particulate matter (PM) emissions

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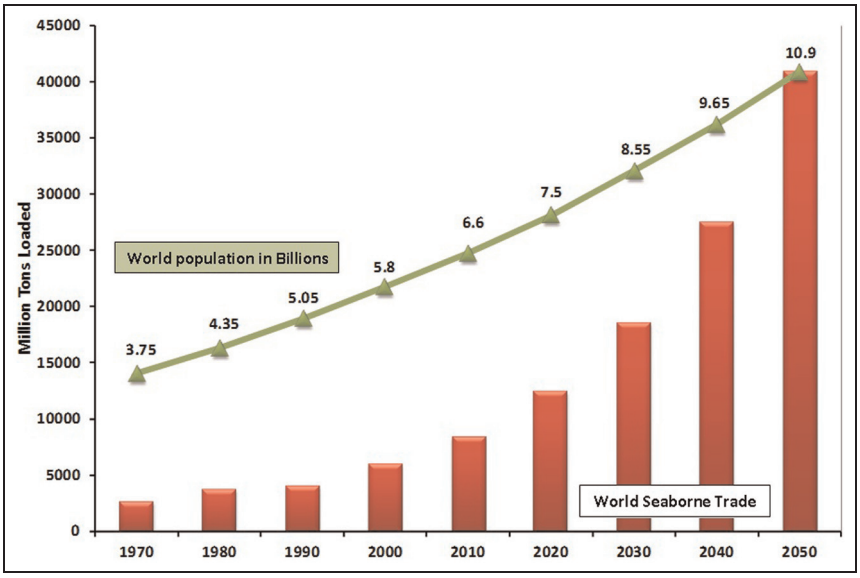


Figure 1. Prospective of world seaborne trade and population growths till 2050.[9,10]

are responsible for 3%–8% of global PM2.5-related mortalities.[4] Moreover, the increase in fuel cost, which represents the main element in the ship's operating cost, especially with the expectation that the prices will continue to increase in the future, has added another difficulty to conventional marine fuels.[5]

Due to incompliance of traditional marine fuels with the latest emission regulations as a result of the continuous increase in fuel prices, the search for alternative marine fuels is deemed of paramount importance in spite of the challenges that they will face. Many researches in this area[6–8] have realized the fact that the selection of an alternative marine fuel is facing the following problems: fuel availability, bunkering operation in ports, storage onboard, ship sailing time and engine room design.

Accordingly, the next sections of this article will illustrate the different types of alternative fuels, which are currently available. Moreover, the properties of the defined alternative fuels will be highlighted. Also, the challenges that face their applicability will be addressed.

The driving forces behind the global energy demand

Population and economic growths are the most important drivers of growing global energy demand because the increase in the world population leads to subsequent increase in the development of world economy and seaborne trade. Figure 1 shows the average population growth against the seaborne trade assuming that the population growth is increasing by 1% each year over the period 2012–2050 and an average output growth for world seaborne trade is increasing by 4% over the same period.[9,10]

It could be predicted from Figure 1 that the global seaborne trade will almost double by 2030 compared to 2010. This is expected to lead to changes in the shipbuilding market trend (e.g. increase in the number of ships and increase in ship speed and size), hence increasing the demand for liquid fuels. Moreover, International Energy Outlook 2013 expects that over the next three decades, global energy consumption (GEC) of liquid fuels in the transportation sector will grow from 96 (in 2010) to 131 (in 2040) quadrillion BTU (see Figure 2).[11]

Several models that estimate the oil production have shown that there will be a peak when about half of the original resources have been extracted, which will then be followed by a rapid decline. Therefore, oil analysts proposed some solutions to offset such decline in the production of oil:[12,13]

1. Improvements in oil recovery factors due to the use of tertiary recovery methods (enhanced recovery oil).
2. Use of nonconventional fuels (e.g. liquefied natural gas (LNG), heavy oils, tar sands, shale oils and biofuels).
3. Expand the share of renewable energies (solar–wind–biomass, etc.).
4. Improvement in the end-use efficiency (facing the problem from demand not from supply).

Marine propulsion systems differ from those used in land-based transportation with regard to their size and range of power, hence only certain types of alternative fuels can be used as bunker fuels aboard ships.

Current situation of ships' emission regulations

Due to the fact that more than 50% of a ship's operating expenses is generally the fuel oil cost, most of the shipowners use heavy fuel oil (HFO) because of its

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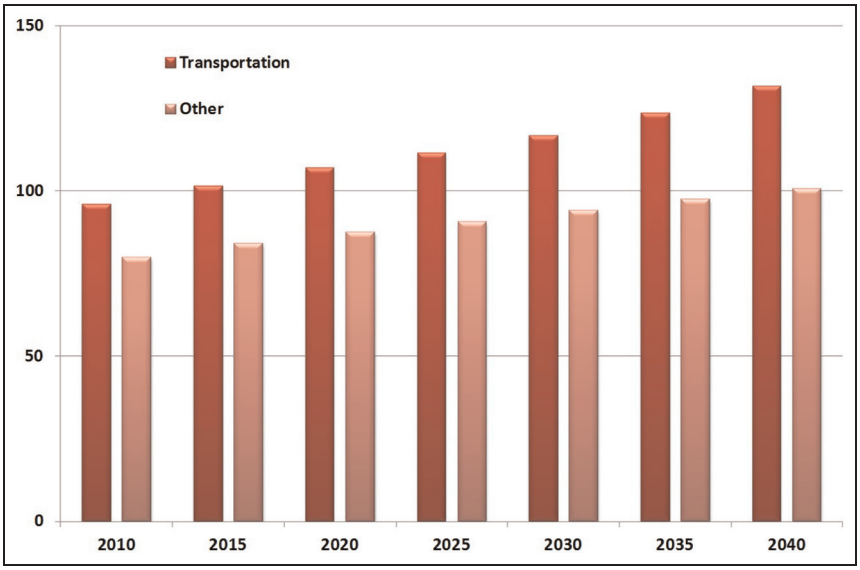


Figure 2. World's liquid consumption by end-use sector, 2010–2040.[11]

economy and availability, and regardless its technical problems that limit the use of HFO of high sulfur content as well as its bad impact on the marine environment.

IMO has introduced a regulation to limit the emissions from marine engines; last May 2005 was the starting date for implementation of the provision of International Convention for the Prevention of Pollution from Ships (MARPOL) 73/78/97 convention, which aims to reduce air pollution from ships, specifically NO_x and SO_x.

The current allowable NO_x emission level according to IMO regulation depends on the speed category of the engine and ranges from 17 g/kW·h for engine speed $<$ 130 r/min to 9.984 g/kW·h for engine speed $>$ 2000 r/min. For compliance purposes, all sea-going ships must prepare Engine Air Pollution Prevention (EAPP) and International Air Pollution Prevention (IAPP) certificates for inspection by port-state control.[15]

The IMO and the European Union (EU), the rule suppliers regarding SO_x emission reduction, have moved forward with the issue through a different timetable and on a different geographical scale. The IMO Marine Environment Protection Committee adopted amendments of the International Convention for the Prevention of Pollution from Ships (MARPOL) Annex VI regulations on sulfur oxide requiring a maximum 3.50% content by 1 January 2012 and 0.50% by 1 January 2020 globally.[16] Moreover, Annex VI also imposed a 1.5% sulfur limit on marine fuels in Emission Control Areas (ECAs) effective since May 2006. That limit was reduced to 1.0% sulfur effective from 1 July 2010 and will be further reduced to 0.1% sulfur beginning January 2015.[17] Figure 3 shows the existing and potential new ECAs.

There are different methods available for reducing ship emissions in order to comply with the IMO requirements, which have an added cost on the ship.

Table 1. Available methods for reducing ship emissions.[20–23]

Component	Reduction method	Potential reduction
NO_x	Selective catalytic reduction (SCR)	95%
	Emulsification	20%–25%
	Humid air	70%
	Engine tuning	50%–60%
	Exhaust gas recirculation	10%–30%
SO_x	Fuel switching process[a]	60%–90%
	Seawater scrubbing; exhaust below water line	Up to 95%
CO_2	Energy management	1%–10%
PM	Electrostatic filters	Up to 85%

PM: particulate matter.
[a]Switching from residual fuel to distillate fuel.

These methods and their reduction potentials are listed in Table 1. However, the pressure for wider and more severe limits on SO_x, NO_x and particulates will accelerate, especially in developed economies and coastal areas. Some countries have imposed their own regulations to control marine emissions on a local basis as shutting down all engines and connecting to shore supply. EU strategy for controlling air pollution calls all ships in EU ports to burn fuel with maximum sulfur content of 0.1%, which would force ships to carry low-sulfur fuel specified for this purpose.[19]

Alternative fuels

The different alternative fuels that include coal, biodiesel, Fischer–Tropsch (F-T) diesel, alcohol, natural gas and hydrogen will be illustrated related to renewability, adaptability to existing engines, safety, cost, performance, economy and finally to environmental impacts (emissions), with emphasis on the hydrogen and

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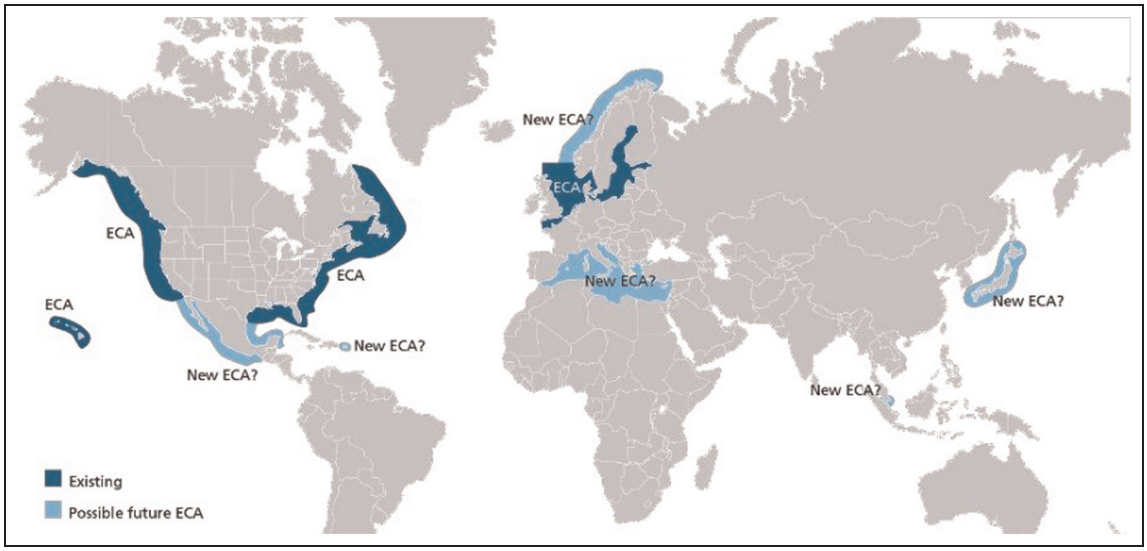


Figure 3. Existing and potential new ECAs around the globe.[18]
ECA: Emission Control Area.

natural gas as they were under thorough investigation in many research centers during the last few years.

1. *Coal*. It has the advantage of lower cost, safe operation, reasonable availability and adaptability as fuel for mechanical stroker–pulverized coal-fluidized bed combustion.[24] But due to its higher sulfur content, it is completely outside IMO regulations. So it considers a nonrenewable energy resource.
2. *Biodiesel*. It is most commonly produced from soybean oil, rapeseed oil, sunflower oil, corn oil and olive oil,[25] along with some wastes, such as used waste frying oils,[26] which appear to be attractive candidates for biodiesel production. Biodiesel is a renewable fuel compatible with the current engines; its use would reduce dependence on fossil fuel and reduce air pollution and related public health risks.[27] The disadvantages of biodiesel are cold weather starting, some storage instability and slight increase in NO_x emissions $(+2: +5\%)$;[28] this increase is due to the higher oxygen content of the fuel. However, some reductions in NO_x emissions can be attained by retarding the timing of ignition and slowing the burn rate of the fuel in the combustion chamber.[28] Despite its presence as environmentally friendly source, its availability is limited.
3. *Fischer–Tropsch (F-T) diesel from natural gas and coal*. It is a production process that produces a diesel fuel from natural gas or coal (gas to liquid (GTL) or coal to liquid (CTL)) via steam reforming, auto-thermal reforming or gasification. Its availability and adaptability will follow the main fuel source (natural gas or coal). F-T diesel fuel has no sulfur, almost no aromatics and high cetane. On the other hand, F-T process is very energy intense and its capital investment is large; this makes it very costly fuel.[29]
4. *Alcohols*. They are of two types, ethanol (C_2H_5OH) and methanol (CH_3OH), which can be produced from sugarcane waste and any many agriculture products (renewable sources). It is not a new idea; it was used in motor vehicle since 1954. The availability and indigenous sources, ease of handling, low emission and high-thermal efficiency obtainable with its use make it as a logical alternative in future, especially to hydrogen generation for fuel cells. Recent studies[30] showed the possibility of using methanol as alternative fuel, especially in dual-fuel engines. The problems associated with the use of methanol or its blends are the emission of aldehyde, phase separation, vapor lock, cold starting and cost-effectiveness.[27]
5. *Natural gas*. It is a mixture of paraffinic hydrocarbons such as methane, ethane, propane and butane. Natural gas is a low-density and low-sulfur content fuel as compared to petroleum products and is practically free from carbon monoxide emission. Natural gas is converted to LNG by cooling it down to -162 °C, at which it becomes a liquid, and this process reduces its volume by a factor of more than 600. Thus, natural gas has emerged as the most preferred fuel due to its inherently environmental benignity, greater efficiency and cost-effectiveness.[31] Figure 4 shows the distribution of proved natural gas reserves in 2012.[32]

The use of natural gas in internal combustion engines has been researched thoroughly to reach the optimum case in both engine performance and environmental impact. Both types of internal combustion engines were studied: the compression ignition and the spark ignition engines. All problems associated with the use of natural gas in these engines were dependent on the injection timing inside the engine cylinders and the cylinder geometry; accurate control is needed to

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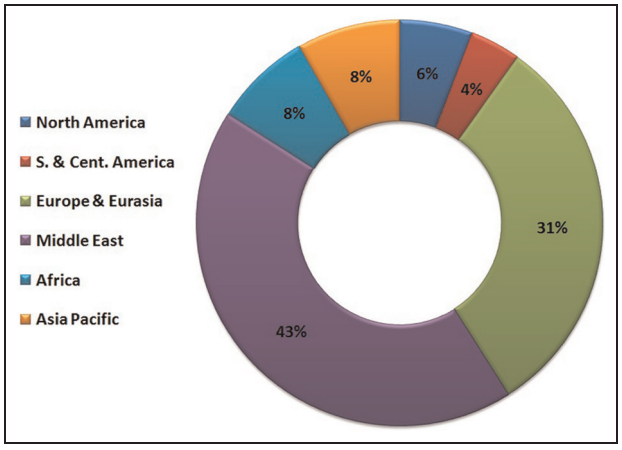


Figure 4. Distribution of proved natural gas reserves at the end of 2012.[32]

avoid engine knocking and high-emission formation levels. Lean burn concepts also were investigated to reach low-emission conditions.[8]

Another application for the natural gas is the gas turbine field. It is used worldwide in electric generation stations either alone or in cogeneration plants, where the resulting heat is used for district heating purposes to increase the plant overall efficiency. Almost all gas turbine manufacturers produce units working with natural gas. In fact, gas turbines are better used with gas rather than liquid fuels for the cleanliness of gas combustion eliminating the soot production and leaving the turbine blades clean of any deposits. From the emission's viewpoint, the natural gas itself produces fewer emissions in gas turbines than in internal combustion engines, except for CO_2 because of the higher fuel consumption in gas turbines due to its lower efficiency than internal combustion engines.[33]

Recently, the marine field witnessed the introduction of natural gas fuel for diesel engines in LNG carriers, where the boil-off gas evaporated from the cargo is used in diesel engines instead of boilers as was the case for decades. Proposals to use gas turbines instead of diesel engines have been introduced, but none was implemented in reality till now.

6. *Hydrogen*. It is considered as a renewable source of energy and has been considered to be the fuel of the future for decades. The scientific research concerning the use of hydrogen in transportation began shortly after the first oil crisis. Many car manufacturers started development programs to produce a car running by hydrogen fuel in internal combustion engines.[34] The beginning was in Germany and Japan and the United States followed them. Nowadays, the United States has national programs for the development of hydrogen systems especially for fuel cell applications to overcome any combustion-related problems.

Combustion of hydrogen inside internal combustion engines has been, and still is, the subject for many research programs in many countries. Like the natural gas, the main problems associated with the application of hydrogen in internal combustion engines include the engine knocking; air–fuel ratio and intake temperature were found to be the main causes for this problem, and their optimization is a must to have a knock-free engine.[15,35]

Regarding the emissions, it is obvious that the hydrogen produces fewer emissions of CO_2, SO_x and PM in both partial-load and full-load engine operations. However, the high-peak combustion temperatures in hydrogen engines usually increase the NO_x emissions.[36] This problem could be controlled using the nowadays well-proven technologies, such as exhaust gas recirculation and selective catalytic reduction (SCR) units.[25] The environmental impact of hydrogen primarily depends on its source and will be positive if hydrogen is produced from clean energy source, through the process of water analysis, using solar cell. However, the impact will be negative when using fossil fuel to produce hydrogen. Despite all the mentioned advantages, using of hydrogen as an alternative fuel is still very limited due to its drawback regarding the economical, adaptability and safety operation issues.

Comparison between alternative marine fuels

All alternative fuels proposed above for marine use have been qualitatively compared by Banawan et al.[8] and Radwan et al.,[37] which reflect, to extended range, the discussion carried out through the previous section, regarding the alternative marine fuel. Table 2 shows the results of this comparison. The comparison revealed that LNG is collectively the best alternative fuel for marine use due to its reasonable cost, acceptable adaptability to existing engines and availability. The only drawback is being nonrenewable compared to the hydrogen, which comes in the second grade after LNG.

The introduction of LNG onboard commercial ships introduces several issues with regard to fuel consumption, cost saving, emission benefits, compliance with the current IMO regulations, gas storage, weight and volume onboard, engine conversions, safety and finally classification's view. In the next section, a comparison between LNG and HFO covering most of these issues, for three different marine diesel engine ratings, will be illustrated and its results will be discussed.

Comparison between HFO and LNG

Fuel consumption and cost saving

LNG cost includes production, liquefaction and logistics and bunkering costs, and therefore, its cost will be different from one area to another. Taking into account

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Table 2. Comparison between alternative fuels for marine use.[37]

	Coal	Biodiesel	F-T diesel	Alcohol	H_2	LNG
Availability	Good	Very good	Good	Very good	Excellent	Very good
Renewability	Fairly good	Good	Good	Very good	Excellent	Fairly good
Safety	Excellent	Excellent	Excellent	Very good	Fairly good	Excellent
Cost	Excellent	Good	Good	Good	Fairly good	Excellent
Adaptability	Good	Excellent	Excellent	Good	Good	Excellent
Performance	Good	Very good	Very good	Good	Good	Excellent
Environmental impact	Bad	Good	Very good	Good	Excellent	Excellent

F-T: Fischer–Tropsch; LNG: liquefied natural gas.

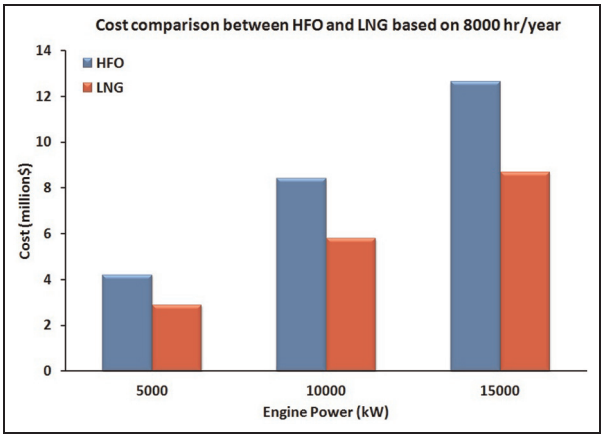


Figure 5. Cost comparison between HFO and LNG for different engine ratings.
HFO: heavy fuel oil; LNG: liquefied natural gas.

the lower heating value (LHV) of fuels, the price of LNG is cheaper than HFO. The basic assumption for the fuel price scenario is a continuous price increase due to expected increase in oil and gas production costs. The current fuel price, in 2013, is about 15–16 USD/mmBTU for HFO and about 11–12 USD/mmBTU for LNG.[38] HFO and LNG are expected to follow the same increment price trend in the years to come, and by 2020, the prices are expected to be about 17 and 14.7 USD/mmBTU for HFO and LNG, respectively.[39] Based on typical load for all diesel engines at 85% maximum continuous rating (MCR) and 8000 working hours per year, the fuel saving cost is estimated for three different engine ratings, namely, 5000, 10,000 and 15,000 kW, and the estimations revealed average fuel saving costs of 1.32, 2.63 and 3.95 million USD/year, respectively, as shown in Figure 5. This is equivalent to an annual cost reduction of about 31% at each engine rating.

When studying the benefits of changing from HFO- to LNG-fueled ships, other costs should be taken into consideration such as investment, installation and maintenance costs. According to a previous study, which was carried out by Wärtsilä Company,[40] aimed to evaluate the benefits of changing from HFO-fueled engine (fitted with scrubber system) to LNG-fueled engine, the machinery investment first cost was estimated to be about 14 $/kW for both HFO-fueled engines and LNG-fueled engines. Moreover, the annual machinery costs (including annual capital, lubricating oil, maintenance, scrubber and SCR operating costs) were 2600 and 2100 $/kW for HFO- and LNG-fueled engines, respectively. This reveals that changing from HFO to LNG will result in additional saving from the annual machinery costs by about 500 $/kW. In addition, the increase in sailing time in ECAs, where a high-quality fuel is needed, will add more economic benefits to LNG- than HFO-fueled engines, where more strict emission regulations are implemented.

Environmental benefits

The Environmental Protection Agency (EPA) inventory method, which is used to estimate the quantity of ships' emissions, thereby includes defining the input parameters related to the type of vessel, the type of fuel consumed and the mode of operation. Consequently, the emission quantity can be estimated using the following formula[8,41]

$$\boldsymbol{E} = \boldsymbol{P} \times \boldsymbol{LF} \times \boldsymbol{efd} \times \boldsymbol{t} \quad (1)$$

where "$\boldsymbol{E}$" is the emission quantity (g), "$\boldsymbol{P}$" is the engine power at MCR (kW), "$\boldsymbol{LF}$" is the load factor (%), "$\boldsymbol{efd}$" is the emission factor (g/kW·h) in diesel mode and "$\boldsymbol{t}$" is the engine running time (h).

By substituting the average emission factors of SO_x, NO_x, CO_2 and PM for both HFO and LNG for *two-stroke* diesel engines (data extracted from Banawan et al.,[8] California Air Resources Board[42] and Gerilla et al.[43]) at the same power and running hours, it was possible to estimate the environmental benefits due to shifting from HFO to LNG (*in case of dual-fuel engine*). Figure 6 shows the results of this estimation. It is apparent from these results that the shift from HFO to LNG has resulted in reduction in SO_x, NO_x, CO_2 and PM emissions by approximately 98%, 86%, 11% and 96%, respectively.

From the life-cycle emission's point of view, LNG gives less acidification and eutrophication potentials compared with HFO. In contrast, the use of LNG does not decrease the global warming potential (GWP) by more than 8%–20%, the amount depending mainly on the magnitude of the methane slip from the gas

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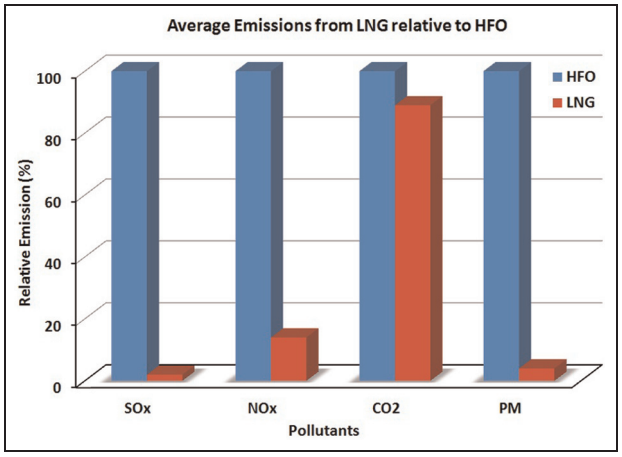


Figure 6. Relative pollutant's emission for LNG and HFO.
LNG: liquefied natural gas; HFO: heavy fuel oil; PM: particulate matter.

Table 3. Standard specifications of LNG bottles.[45]

Cylinder size (m^3)	15	17	19	21	23
Average weight (kg)	66	73	84	92	98

engine.[44] Small leakages of CH_4 can cancel out the beneficial effect on the GWP from the reduced CO_2 emission with LNG. The methane slip during combustion is uncertain and different emission factors were found. In addition, another aspect is that the methane slip from the engine is a technical problem that may be reduced with a catalytic convertor.[44]

Gas storage, weight and volume changes

Natural gas can be stored either as "compressed natural gas" (CNG) or "LNG" according to the allowable weight and volume onboard ship as well as the design and operation of the propulsion system. The relative weight and volume for CNG and LNG compared to fuel oil were calculated based on the following assumptions:

1. CNG density (ρ_{CNG}) is 0.19 tons/m^3 at storage pressure of 250 bar.
2. CNG storage is in a steel bottles connected in a package named bottle battery. The standard size of the steel bottle is given in Table 3.
3. LNG is stored at −162 °C, LNG density (ρ_{LNG}) is 0.450 tons/m^3 at 1 bar, in a cylindrical cryogenic tank with domed ends that typically sustain pressure from 0.3 to 10 bar.
4. LNG tank is made from armed fiber–reinforced plastics with a tensile strength of 1029 MPa and material density of 1.4 tons/m^3.
5. The required tank wall thickness (T) for keeping LNG storage temperature constant at −162 °C can be determined from the following equation

$$T = \frac{E \cdot r \cdot P}{F} \quad (2)$$

where r is the tank radius, P is the operating pressure, F is the tensile strength and E is the safety factor, taken to be = 3.

6. Fuel mass (mf) can be calculated by multiplying a horsepower (hp) by specific fuel consumption (SFC) and the ship endurance (h)

$$mf = SFC \cdot hp \cdot h \quad (3)$$

7. The volume of the fuel (Vf) can be determined by

$$Vf = \frac{mf}{\rho f} \quad (4)$$

The relative weight and volume were calculated, and it was found that the required volume of fuel and tank for 1 m^3 marine fuel oil is equivalent to 2 and 4.5 m^3 for LNG and CNG, respectively; this is mainly due to the difference between fuel densities. The relative weight of the fuel was reduced when using CNG or LNG by 20% compared to marine fuel oil due to the higher NG calorific value. On the other hand, the corresponding weight of CNG tank was found to be triple of LNG. Therefore, for each different application, a feasibility study should be done to figure out the most appropriate storage method.

Conversion of engines

The high cost of vessel conversion or replacement presents a challenge to the long-term return from fuel saving. LNG conversion can cost up to $7 million for a medium-sized tug, almost $11 million to convert a large ro-ro/passenger ferry and up to $24 million to convert a Great Lake bulk carrier. The majority (approximately 80%–85%) of the conversion or incremental cost of a new ship typically comes from the installation of LNG storage tanks and related safety systems, with the remaining related to the conversion of the vessel engines. With payback on the conversion typically over 10 years, not all operators will be willing or able to take on such a long-term investment.[46,47]

The conversion of diesel engines from diesel-fueled to gas-fueled will require certain changes to be performed. These changes include the following: the modification of the main engines, natural gas supply system, supply piping and valves, gas detection units and alarms and exhaust ventilation system, among others. The conversion or replacement cost will depend mainly on the ship type and characteristics.

On the other hand, and in addition to the economic benefit of saving in the fuel cost by using LNG instead of marine diesel oil (MDO), maintenance cost is also considered as a very important aspect that affects economic consideration. It was shown by Banawan et al.[8]

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that the mean time between maintenance (MTBM) for natural gas–fueled engines could be increased by three to four times over diesel-fueled engines. This means that the total running hours of natural gas–fueled engines will be at least three times those of engines working by diesel oil. For other maintenance activities that include routine maintenance, lubricating oil consumption and spare parts consumptions, the cost is expected to be reduced in case of using natural gas by about 50% of original cost.[48,49] Technically, the conversion process from MDO to dual-fuel natural gas engines leads to that the engine acts according to the Otto principle. As the fuel is mixed with the air before compression starts, a gas pressure of about 5 bar is sufficient. Close to top-dead center, a very small amount of MDO is injected in order to trigger ignition. Provided that an adequate gas supply system is installed, dual-fuel engines can accept all qualities seen in LNG shipping. In addition to running on gas, dual-fuel engines can run on MDO. In this case, the dual-fuel engine acts as the normal diesel engine.[20]

Rules and regulations related to the use of LNG as a ship fuel

Due to the significant difference in properties, characteristics and behavior between LNG and conventional marine fuels, many safety, technical, operational and training aspects should be considered and due attention should be paid in order to ensure safe storage, transport, bunkering and use of LNG as ship fuel and to prevent leakage of LNG liquid or vapor and control all sources of ignition. All these aspects should be covered by both International and National Regulatory Frameworks. Among the most relevant regulatory and standardization bodies that are involved in this process are the IMO, the International Organization for Standardization (ISO), the International Electrotechnical Commission (IEC), the Society of International Gas Tanker & Terminal Operators (SIGTTO), the National Fire Protection Association (NFPA), the American Petroleum Institute (API) and a number of Classification Societies, which are members in the International Association of Classification Societies (IACS).

IMO adopts its own regulations for maritime safety and security, efficiency of navigation and prevention and control of pollution from ships. As a first contribution, the IMO has addressed the use of LNG as fuel in its International Code for Construction and Equipment of Ships Carrying Liquefied Gases in Bulk (IGC Code),[50] which is applicable to liquefied gas carriers and covers the use of the boil-off gas as fuel. On June 2009, and as a second contribution, the IMO had adopted the "Interim guidelines on safety for natural gas fuelled engine installations in ships," by its resolution MSC.285(86),[51] and provided criteria for the arrangement and installation of machinery for propulsion and auxiliary purposes, using natural gas as fuel for ships other than those covered by the IGC Code. Currently, the IMO is continuing its work toward the development of the "International Code of Safety for Gas Fuelled Ships" (IGF Code), which agreed to rename it to "International Code of Safety for Ships Using Gases or Other Low Flashpoint Fuels" and is expected to be finalized in 2015.[52]

ISO works closely with IMO to develop standards for the shipping industries. In the area of using LNG as ship fuel, ISO has developed and is still currently working on the development of a number of standards. Among these important standards, one should mention the standard for "Installation and equipment for liquefied natural gas—ship-to-shore interface and port operations" (ISO 28460:2010),[53] which includes the requirements for ship, terminal as well as port service providers that ensure safe transit of an LNG carrier through the port area and safe and efficient transfer of its cargo. Very recently (June 2013), the International Association of Oil and Gas Producers (OGP) in association with ISO has developed "Guidelines for systems and installations for supply of LNG as fuel to ships" (ISO/DTS 18683) and also references to ISO/TC 67 WG 10.[54]

IEC prepares and publishes international standards related to electrical, electronic and related technologies not covered by the ISO. There are some IEC Standards that are related to the deal with ships carrying dangerous/hazardous materials such as LNG. Among these standards, one could consider the following: the IEC 60092-502 Standard: "Electrical installations in ships—part 502: tankers—special features" that deals with the electrical installations in tankers carrying liquids that are flammable, either inherently or due to their reaction with other substances or flammable liquefied gases;[55] the IEC 60079 Standard: "Electrical apparatus for explosive gas atmospheres"[56] and the IEC 61508 Standard: "Functional safety of electrical/electronic/programmable electronic safety-related systems."[57]

SIGTTO specifies and promotes Standards & Best Practice for Liquefied Gas Industries. Among their important guidelines that could be related to the use of LNG as ship fuel are the following: the LNG Ship-to-Ship Transfer Guidelines, which include guidance for safety, communication, maneuvering, mooring and equipment for vessels undertaking side-by-side ship-to-ship transfer, and the Liquefied Gas Fire Hazard Management, which includes the principles of liquefied gas fire prevention and fire fighting.[58]

NFPA provides and advocates codes and standards, research, training and education related to the risks of fire and other related hazards. There are many important publications that could be related to the use of LNG as ship fuel. Among these publications, one could consider the following: NFPA 59A (Standard for the Production, Storage and Handling of LNG) and NFPA 302 (Fire Protection Standard for Pleasure and Commercial Motor Craft), which establishes minimum

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requirements for the fire protection and prevention in machinery spaces onboard boats of less than 300 gross tons.[52]

API produces standards, recommended practices, specifications, codes and technical publications that cover each segment of the industry. Most of the standards and recommended practices are dedicated to a single type of equipment. Among these publications that could be applicable to the use of LNG as ship fuel are the following: API RP 521—"Guide for Pressure-relieving and Depressuring Systems Petroleum, petrochemical and natural gas industries," API Std 617—"Axial and Centrifugal Compressors and Expander-compressors for Petroleum, Chemical and Gas Industry Services" and API Std 620—"Design and Construction of Large, Welded, Low-Pressure Storage Tanks," which includes requirements for refining and storage tanks.[59]

Most of the 13 IACS members' Classification Societies have developed guidelines for the use of LNG as ship fuel within their class rules. These guidelines are based on the IMO Interim Guidelines MSC.285(86), with additional specific requirements for each class, and give guidance for the design, construction and operation of LNG-fueled ships. It should be noted here that these guidelines are not obligatory, and each flag state must agree upon the operation of LNG-fueled vessels sailing in their National Waterways. Both Germanischer Lloyd (GL) and Det Norske Veritas (DNV) have been involved in many national and international research projects investigating different aspects of using LNG as ship fuel, and they could be considered as pioneers in this active area of research (guidelines were first published in 2010).[60,61]

Conclusion

The work presented in this article highlighted the difficulties that the current liquid fuels face, which are in use nowadays with marine diesel engines. These included the difficulty of compliance with the new emission regulations set out in Annex VI, 2008 amendments of the MARPOL Convention, and the continuous increase in the fuel cost. In addition, more restrictive regional/global emission standards are expected to be forced by the regulators, in the nearest future. While searching for alternative marine fuels that can conquer these difficulties, it is found that LNG could offer significant advantages over the current marine fuel (HFO) in terms of the fuel cost by about 31% per year. It is also showed that LNG has environmental benefits through an average reduction of SO_x, NO_x, CO_2 and PM pollutants by an amount of about 98%, 86%, 11% and 96%, respectively. Due to lack of sulfur content, the maintenance cost will be decreased and the time between overhauls will be increased. The emergence of LNG as an alternative marine bunker fuel will depend on the regulatory authorities with regard to emission reduction and availability of ports' infrastructure and logistic for refueling with LNG. Moreover, the wide adaptation of using LNG as fuel onboard will increase the motivation to fuel cell and gas turbine as main prime movers. It is also expected that the usage of LNG will offset the point of conventional oil depletion to some extent. It should be noted that most of the current marine diesel engines can meet the restricted emission regulations with some economic penalties. Due to the fact that LNG has significant differences in terms of properties, characteristics and behavior than conventional marine fuels, due attention should be paid in order to ensure safe storage, transport, bunkering and use. Before using LNG as marine fuel, all these aspects should be enclosed by both National and International Regulatory Frameworks. Among the most relevant regulatory and standardization bodies that are currently active in this process are IMO, ISO, IEC, SIGTTO, NFPA, API and a number of Classification Societies, which are members in the IACS.

Declaration of conflicting interests

The authors declare that there is no conflict of interest.

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References

1. Kołwzan K and Narewski M. Study on alternative fuels for marine applications. *Clean Shipp Curr* 2012; 1(3): 1–43.
2. Eyring V, Isaksen Ivar SA, Berntsen T, et al. Transport impacts on atmosphere and climate: shipping. *Atmos Environ* 2010; 44(37): 4735–4771.
3. Norwegian Maritime Technology Research Institute. Second IMO GHG Study 2009. Technical report, http://www.imo.org/blast/blastDataHelper.asp?data_id=27795&filename=GHGStudyFINAL.pdf (2009).
4. Eide MS, Dalsøren SB, Endresen Ø, et al. Reducing CO_2 from shipping—do non-CO_2 effects matter? *Atmos Chem Phys* 2013; 13: 4183–4201.
5. Bui Y. Machinery concepts and LNG for meeting IMO Tier III rules. *Wartsila Tech J*: InDetail 2011; 1:31–38.
6. Levander O. Handy size LNG cruise ship concept. *Wartsila Tech J: InDetail* 2011; 1:44–51.
7. Stuer-Lauridsen F, Odgaard T, Winter Graugaard C, et al. *Natural gas for ship propulsion in Denmark: possibilities for using LNG and CNG on ferry and cargo routes*. Danish Ministry of the Environment; Environmental Protection Agency (EPA). Project no. 1338, October 2010.
8. Banawan AA, El-Gohary MM and Sadek IS. Environmental and economical benefits of changing from marine diesel oil to natural-gas fuel for short-voyage high-power passenger ships. *Proc IMechE Part M: J Engineering for the Maritime Environment* 2010; 224(2): 103–113.
9. United Nations—Department of Economic and Social Affairs—Population Division. *World population prospects—the 2012 revision, key findings and advance tables.*

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New York: United Nations—Department of Economic and Social Affairs—Population Division, 2013.
10. United Nations Conference on Trade and Development (UNCTAD). *Review of maritime transport*. New York; Geneva: UNCTAD, 2012.
11. U.S. Energy Information Administration (EIA). *International energy outlook 2013*. DOE/EIA-0484, July 2013. Washington, DC: EIA.
12. Bentley RW. Global oil and gas depletion: an overview. *Energ Policy* 2002; 30(3): 189–205.
13. Goldemberg J. The promise of clean energy. *Energ Policy* 2006; 34(15): 2185–2190.
14. Thomas RF. Fuel quality related to un-seaworthiness. *Trans Inst Mar Eng C* 1996; 108: 81–84.
15. Lin B and Lin CY. Compliance with international emission regulations: reducing the air pollution from merchant vessels. *Mar Policy* 2006; 30(3): 220–225.
16. Gritsenko D and Yliskylä-Peuralahti J. Governing shipping externalities: Baltic ports in the process of SO_x emission reduction. *Marit Stud* 2013; 12: 10.
17. Frederick A and Amokeye A. LNG as marine fuel. In: *The 17th international conference and exhibition on liquefied natural gas*, Gas Technology Institute (GTI). Energy Learning Center: Houston, TX, 16–19 April 2013.
18. DNV Report, Joint Industry Project. LNG fuel bunkering in Australia: infrastructure and regulations. Public version of partners' internal report, project no. PP033420, report no. CTC_R_2012038, DET NORSKE VERITAS PTE LTD, Clean Technology Centre, 20 January 2013.
19. Markku, M. International and EU regulations with regards to lower sulphur emissions: EMSA's involvement in promoting the use of LNG as fuel. European Harbour Masters' Committee (EHMC) seminar, Bremen, April 17th, 2013
20. Woodyard D. *Pounder's marine diesel engines and gas turbine*. 8th ed.Oxford: Butterworth-Heinemann, 2004.
21. EPA. Regulatory impact analysis: control of emissions of air pollution from category 3 marine diesel engines. Chapter 4: technological feasibility, EPA, U.S. Environmental Protection Agency (EPA), USA. December 2009.
22. Buhaug Ø, Corbett JJ, Endresen Ø, et al. *Second IMO GHG Study*. London: International Maritime Organization (IMO), 2009.
23. Ariana IM, Nishida O, Fujita H, et al. *Removal of marine diesel particulate matter by electrostatic precipitator*. ICESP X—Australia, paper 4A2, http://www.isesp.org/ICESPXPAPERS/PDFS/Paper4A2.pdf (2006).
24. El-Wakil MM. Powerplant technology. 2nd ed. New York: McGraw-Hill, Inc., 1988.
25. Kalligeros S, Zannikos F, Stournas S, et al. An investigation of using biodiesel/marine diesel blends on the performance of a stationary diesel engine. *Biomass Bioenerg* 2003; 24(2): 141–149.
26. Yu CW, Bari S and Ameen A. A comparison of combustion characteristics of waste cooking oil with diesel as fuel in a direct injection diesel engine. *Proc IMechE Part D: J Automobile Engineering* 2002; 216(3): 237–243.
27. MacLean HL and Lave LB. Evaluating automobile fuel/propulsion system technologies. *Prog Energ Combust* 2003; 29(1): 1–69.
28. Randall von Wedel. Technical Handbook for Marine Biodiesel in Recreational Boats. CytoCulture International, Inc. Point Richmond, CA. Second Edition. April 22, 1999. Marine Biodiesel and Education Project for San Francisco Bay and Northern California. Prepared for the National Renewable Energy Laboratory. U.S. Department of Energy.
29. Gielen D and Unander F. Alternative fuels: an energy technology perspective. Head of Publications Service, http://s3.amazonaws.com/zanran_storage/www.iea.org/ContentPages/26160780.pdf (2005).
30. Per Westling. Methanol- A good alternative for ferries and short sea shipping. In: Interferry 38th annual conference, Malta, 5–9 October 2013.
31. Kumar S, Kwon HT, Choi KH, et al. LNG: an eco-friendly cryogenic fuel for sustainable development. *Appl Energ* 2011; 88(12): 4264–4273.
32. British Petroleum (BP). Statistical Review of World Energy, www.bp.com/content/dam/bp/pdf/statistical-review/statistical_review_of_world_energy_2013.pdf (2013, accessed 5 August 2013).
33. El-Gohary MM. The future of natural gas as a fuel in marine gas turbine for LNG carriers. *Proc IMechE Part M: J Engineering for the Maritime Environment* 2012; 226(4): 371–377.
34. Morsy El Gohary, M. and El-Sherif, H.A., 2006. Future of hydrogen as green energy in marine application. 9th World Renewable Energy Congress. Florence, Italy. August 2006, Elsevier.
35. El-Gohary MM and Saddiek I. Utilization of alternative marine fuels for gas turbine power plant onboard ships. *JNAOE* 2013; 5: 21–32.
36. El-Gohary MM. Overview of past, present and future marine power plants. *J Mar Sci Appl* 2013; 12: 219–227.
37. Radwan A, Morsy M and Elbadan A. Economical and environmental advantages of using NG as a fuel in inland water transport. In: *25th CIMAC conference*, CIMAC. International Council on Combustion Engines: Vienna, 2007, paper no. 114.
38. Harsema A. LNG fuel tank container: a new approach to LNG bunkering. Chicago, IL, September, 2013. Marine Service GmbH, http://hhpsummit.com/pdfs/HHP2013 Presentations/BO2-1Ma/3Harsema-Mensonides.pdf
39. Andersen ML, Clausen NB and Sames PC. *Costs and benefits of LNG ship fuel for container vessels*. Hamburg: Germanischer Lloyd, 2011.
40. Levander O. Dual fuel engines latest developments. Technical report, Wärtsilä Company, Concept design, Hamburg, 27 September, http://www.ship-efficiency.org/onTEAM/pdf/PPTLevander.pdf (2011, accessed 20 December 2013).
41. SENES Consultants Limited and Air Improvement Resource, Inc. Review of methods used in calculating marine vessel emission inventories. Report prepared for Environment Canada, Pollution Data Branch, Gatineau, QC, Canada, March 2004.
42. California Air Resources Board. Emissions estimation methodology for ocean-going vessels, http://www.arb.ca.gov/regact/marine2005/appd.pdf (2005).
43. Gerilla P, Kardi T and Kazunori H. Environmental assessment of international transportation of products. *J East Asia Soc Transp Stud* 2005; 6: 3167–3182.
44. Bengtsson S, Andersson K and Fridell E. A comparative life cycle assessment of marine fuels liquefied natural gas. *Proc IMechE Part M: J Engineering for the Maritime Environment* 2011; 225(2): 97–110.
45. Tawfek A. Environmental aspects of natural gas application in inland water transportation. *Alex Eng J* 2005; 44(5): 705–713.

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46. Balaon T, Lowell D, Curry T, et al. Natural gas for marine vessels. Technical Report, American Clean Skies Foundation, Washington, DC, April 2012.
47. Deal AL. Liquefied natural gas as marine fuel a closer look at Tote's containership projects. NEPI working paper, National Energy Policy Institute, USA, 13 June 2013.
48. Nylund I. Field experience with the Wärtsilä 50 dual–fuel engine. In: Proceedings of the 25th CIMAC World Congress on Combustion Engines, Vienna, Austria, 21–24 May 2007. paper no. 239.
49. Levander O and Sipila T. LNG auxiliary power in port for container vessels. *Wartsila Tech J: InDetail* 2008; 2:42–47.
50. International Maritime Organization (IMO). *Draft revised International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk—(IGC Code)*. SUB-COMMITTEE ON BULK LIQUIDS AND GASES, 17th session, Agenda item 9, IMO, 6 June 2012.
51. International Maritime Organization (IMO). *Interim guidelines on safety for natural gas fuelled engine installations in ships*. Annex 11, RESOLUTION MSC.285(86), IMO, Adopted on 1 June 2009.
52. Germanischer Lloyd (GL). Study on standards and rules for bunkering of gas-fuelled ships. Final report prepared by GL for the European Maritime Safety Agency (EMSA). GL, Report no. 2012.005, Version 1.1, March 2013.
53. ISO 28460:2010. Petroleum and natural gas industries—installation and equipment for liquefied natural gas—ship-to-shore interface and port operations. 1st ed.
54. International Association of Oil and Gas Producers (OGP). *Guidelines for systems and installations for supply of LNG as fuel to ships*. Interntional Association of Oil and Gas Producers, London, UK. OGP Draft 118683, 4 June 2013.
55. IEC 60092-502:1999. Electrical installations in ships—part 502: tankers—special features. 5th ed., February 1999.
56. IEC 60079:2012. Electrical apparatus for explosive gas atmospheres (36 parts; Available as BS EN 60079 from BSI, Milton Keynes, UK, or from the IEC, Geneva).
57. IEC 61508:2005. Functional safety of electrical/electronic/programmable electronic safety-related systems (7 parts; Available as BS EN 61508 from BSI, Milton Keynes, UK, or from the IEC, Geneva).
58. SIGTTO. *Ship to ship transfer guide (liquefied gases)*. 2nd ed. London, Witherby & Co. Ltd, 1995. (ISBN 1-85609 082 5)
59. Det Norske Veritas (DNV). *Modalities for the provisioning of LNG as shipping fuel in Flemish Ports. Part II: legal and regulatory*. Report no. /DNV REG no.: 13YB7C6–3, REV 2, 10 July 2012.
60. Det Norske Veritas (DNV). DNV rules for classification of ships/high speed, light craft and naval surface craft. Part 6/Chapter 13: gas fuelled ship installations, DNV, Høvik, July 2013.
61. Germanischer Lloyd (GL). GL rules and guidelines. Section VI: additional rules and guidelines. Part 3/Chapter 1: guidelines for the use of gas as fuel for ships, GL, Hamburg, May 2010.

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